Sr no.	Protocol	Description	Port Number	Encryption	Security Vulnerabilities	Typical Use Cases	Vendors Using It
1	BACnet	A protocol for building automation and control networks.	TCP: 47808	Secure Authentication and TLS encryption	Vulnerable to unauthorized access, DoS attacks	HVAC systems, lighting control, energy management	Honeywell, Johnson Controls, Siemens
2	BACnet/IP	A variant of BACnet protocol using IP networks for building automation.	UDP: 47808	Secure Authentication and TLS encryption	Vulnerable to unauthorized access, DoS attacks	Building automation, control and monitoring	Honeywell, Johnson Controls, Siemens
3	BACnet/IPv6	A variant of BACnet protocol using IPv6 for building automation.	UDP: 47808	Secure Authentication and TLS encryption	Vulnerable to unauthorized access, DoS attacks	Building automation, control and monitoring with IPv6	Honeywell, Johnson Controls, Siemens
4	CANbus	A bus standard for communication in vehicle systems.	N/A	Not Available (Plain Text)	Vulnerable to spoofing, replay attacks	Automotive systems, control units	Bosch, Continental, Delphi
5	CANopen	A higher-layer protocol based on CANbus for industrial automation.	N/A	Not Available (Plain Text)	Vulnerable to unauthorized access, message injection	Industrial automation, motion control systems	Beckhoff Automation, Bosch, Omron
6	CC-Link	A fieldbus protocol for industrial automation in Asia.	TCP: 5000	AES encryption (Inherent encryption)	Lack of authentication, susceptible to eavesdropping	Industrial automation, motion control systems in Asia	Mitsubishi Electric, Panasonic, Hitachi
7	CC-Link IE	An industrial Ethernet protocol for automation systems in Asia.	TCP: 44818,2222	IPsec encryption (Achieved through IPsec implementation)	Potential vulnerabilities in authentication and encryption	Industrial automation, motion control systems in Asia	Mitsubishi Electric, Panasonic, Hitachi
8	CIP	A protocol for communication in industrial automation networks.	TCP/UDP: 44818, 2222	TLS encryption	Potential vulnerabilities in authentication and encryption	Integration of control systems, data exchange, safety devices	Rockwell Automation, Schneider Electric
9	CIP Motion	A protocol for motion control in industrial automation systems.	TCP: 44818, 2222	TLS encryption	Potential vulnerabilities in authentication and encryption	Industrial motion control systems	Rockwell Automation, Schneider Electric
10	CIP Safety	A safety protocol for communication in industrial control systems.	TCP: 44818, 2222	TLS encryption	Potential vulnerabilities in authentication and encryption	Safety-critical applications, control system integration	Rockwell Automation, Schneider Electric
11	DeviceNet	A network protocol for communication with industrial devices.	TCP: 44818, 2222	IPsec encryption (Achieved through IPsec implementation)	Potential vulnerabilities in authentication and encryption	Industrial device communication, sensor integration	Rockwell Automation, Schneider Electric
12	DNP3	A robust and secure protocol for communication in electric power systems.	TCP: 20000-20005	Secure Authentication	Vulnerable to man-in-the-middle attacks, lack of key management	Electric power systems, water/wastewater management	General Electric, Siemens, ABB
13	EnOcean	A wireless communication protocol for energy harvesting devices.	Various	AES encryption (Inherent encryption)	Potential vulnerabilities in authentication and encryption	Energy harvesting devices, building automation, wireless sensor networks	EnOcean, Texas Instruments, Siemens
14	EtherCAT	A real-time Ethernet protocol for communication in motion control systems.	TCP/UDP: 37980	IPsec encryption (Achieved through IPsec implementation)	Vulnerabilities in authentication, data integrity	Motion control, automation systems	Beckhoff Automation, Omron, Bosch
15	EtherCAT P	A power-over-EtherCAT protocol for communication and power delivery.	UDP: 8899	IPsec encryption (Achieved through IPsec implementation)	Vulnerabilities in authentication, data integrity	Motion control, automation systems with power delivery	Beckhoff Automation, Omron, Bosch
16	EtherNet/IP	An industrial Ethernet protocol for real-time control and data exchange.	TCP: 44818, UDP: 2222	IPsec encryption (Achieved through IPsec implementation)	Potential vulnerabilities in authentication and encryption	Integration of control systems, safety devices, data exchange	Rockwell Automation, Schneider Electric
17	EtherNet/IPTap	A protocol for network traffic monitoring in EtherNet/IP networks.	TCP: 2222	IPsec encryption (Achieved through IPsec implementation)	Vulnerable to unauthorized access, data integrity issues	Network traffic monitoring, diagnostics in EtherNet/IP networks	Rockwell Automation, Schneider Electric
18	Fieldbus HSE	A high-speed Ethernet protocol for fieldbus communication.	TCP: 2222	Not Available (Plain Text)	Vulnerabilities in authentication, data integrity	Fieldbus communication, high-speed data exchange	ABB, Emerson, Yokogawa
19	FL Net	FL-net is used in industrial control systems for factory automation.	UDP 55000-55003	Not Available (Plain Text)	Vulnerabilities in authentication, data privacy	Manufacturing, process control applications, Robotics	Omron, Mitsubishi
20	Foundation Fieldbus	A digital communication protocol for process control systems.	TCP: 2222	Not Available (Plain Text)	Vulnerabilities in authentication, data integrity	Process control, monitoring and diagnostics	ABB, Emerson, Yokogawa
21	FOUNDATION FIELDBUS HSE	A high-speed Ethernet protocol for process control systems.	TCP/UDP: 1089-1091	Not Available (Plain Text)	Vulnerabilities in authentication, data integrity	Process control, high-speed data exchange	Emerson, Yokogawa, ABB

Sr no.	Protocol	Description	Port Number	Encryption	Security Vulnerabilities	Typical Use Cases	Vendors Using It
22	H1/H2 Fieldbus	A fieldbus protocol used in process automation and control systems.	TCP: 102	Not Available (Plain Text)	Vulnerable to unauthorized access, data integrity issues	Process automation, control and monitoring	Yokogawa, Honeywell, ABB
23	HART	A protocol for communication with intelligent field devices.	TCP: 5094	Not Available (Plain Text)	Vulnerable to spoofing, tampering	Industrial process monitoring and control	Emerson, Honeywell, Yokogawa
24	HART-IP	A variant of HART protocol using IP networks for industrial applications.	UDP: 5094	IPsec encryption (Achieved through IPsec implementation)	Vulnerable to spoofing, tampering	Industrial process monitoring and control over IP networks	Emerson, Honeywell, Yokogawa
25	ICCP/TASE.2	A protocol for real-time information exchange between control centers.	TCP: 102	TLS encryption	Vulnerable to unauthorized access, data integrity issues	Inter-control center communication, energy management systems	General Electric, Siemens, ABB
26	IEC 60870-5	A protocol for communication in electrical utility automation systems.	TCP: 2404	Not Available (Plain Text)	Lack of authentication, vulnerable to DoS attacks	Monitoring and control of electrical power systems	Siemens, ABB, Schneider Electric
27	IEC 61850	A protocol for communication in substation automation systems.	TCP: 102, UDP: 102	TLS encryption	Vulnerabilities in authentication, data integrity	Electric power substation automation, smart grid applications	ABB, Siemens, Schneider Electric
28	IEC 61883	A protocol for audio and video transmission in professional applications.	UDP: 61883	Not Available (Plain Text)	Vulnerabilities in authentication, data privacy	Audio/video transmission, professional multimedia applications	Sony, Panasonic, Canon
29	IEC 62351	A suite of protocols for secure communication in power systems.	TCP: 102	TLS encryption	Vulnerabilities in authentication, key management	Secure communication in electric power systems	Siemens, ABB, Schneider Electric
30	IEEE C37.118	Defines a method for exchange of synchronized phasor measurement data between power system equipments.	TCP : 4712, UDP : 4713	Not Available (Plain Text)	Vulnerabilities in authentication, data privacy	Energy and Power Industries.	GE, Schneider, ABB, Siemens, SEL, Alstom, etc.
31	J1939	A protocol for communication in heavy-duty vehicles and equipment.	N/A	Not Available (Plain Text)	Vulnerable to spoofing, replay attacks	Heavy-duty vehicle communication, diagnostics and control	Bosch, Cummins, Volvo
32	KNX	A protocol for building automation and control networks.	TCP/UDP: 3671,3672	Secure Authentication and TLS encryption	Potential vulnerabilities in authentication and encryption	Building automation, lighting control, HVAC systems	ABB, Schneider Electric, Siemens
33	KNXnet/IP	A variant of KNX protocol using IP networks for building automation.	TCP: 3671, 3672	Secure Authentication and TLS encryption	Potential vulnerabilities in authentication and encryption	Building automation, lighting control, HVAC systems with IP connectivity	ABB, Schneider Electric, Siemens
34	LonWorks	A protocol for control networks used in building automation.	TCP: 1626	AES encryption (Inherent encryption)	Vulnerabilities in authentication, data privacy	Building automation, lighting control, energy management	Echelon, Siemens, Schneider Electric
35	M-Bus	A protocol for remote reading of utility meters.	TCP: 50000	Not Available (Plain Text)	Vulnerable to unauthorized access, data integrity issues	Utility metering, remote meter reading	Kamstrup, Itron, Siemens
36	MelsecNet	A protocol for communication in Mitsubishi Electric PLC systems.	TCP: 5007	Not Available (Plain Text)	Vulnerabilities in authentication, data integrity	Industrial automation, process control systems	Mitsubishi Electric, Omron, Hitachi
37	MMS	A protocol for real-time data communication in industrial systems.	TCP/UDP: 102	TLS encryption	Vulnerabilities in authentication, data integrity	Industrial control systems, real-time data exchange	General Electric, Schneider Electric, ABB
38	Modbus	A serial communication protocol widely used in industrial automation.	TCP: 502	Not Available (Plain Text)	Lack of authentication, susceptible to eavesdropping	SCADA systems, industrial control and monitoring	Schneider Electric, Siemens, ABB
39	Modbus TCP/IP	A variant of Modbus protocol using TCP/IP for communication.	TCP: 502	Secure Authentication and TLS encryption	Lack of authentication, susceptible to eavesdropping	SCADA systems, industrial control and monitoring	Schneider Electric, Siemens, ABB
40	Modbus/TCP	A variant of Modbus protocol using TCP/IP for communication.	TCP: 502	Secure Authentication and TLS encryption	Lack of authentication, susceptible to eavesdropping	SCADA systems, industrial control and monitoring	Schneider Electric, Siemens, ABB
41	MQTT	A lightweight messaging protocol for IoT and M2M communication.	TCP: 1883	TLS encryption	Vulnerabilities in authentication, data privacy	IoT, remote monitoring, real-time data exchange	IBM, Microsoft, Amazon Web Services
42	OPC DA	A standard for interoperability between industrial automation systems.	TCP: 1024-65535	Not Available (Plain Text)	Vulnerable to unauthorized access, lack of data integrity	Industrial automation, device and software integration	Rockwell Automation, Honeywell, Yokogawa

Sr no.	Protocol	Description	Port Number	Encryption	Security Vulnerabilities	Typical Use Cases	Vendors Using It
43	OPC UA	A standard for interoperability between industrial automation systems.	TCP: 4840	TLS encryption	Vulnerable to unauthorized access, lack of data integrity	Industrial automation, device and software integration	Rockwell Automation, Honeywell, Yokogawa
44	PROFIBUS	A fieldbus protocol for communication in automation systems.	TCP: 3668	Not Available (Plain Text)	Vulnerable to eavesdropping, unauthorized access	Sensors, actuators, controllers in manufacturing	Siemens, Phoenix Contact, ABB
45	Profinet	A communication protocol for real-time data exchange in industrial automation.	TCP/UDP: 34962-34964	IPsec encryption (Achieved through IPsec implementation)	Vulnerabilities in access control, authentication mechanisms	Manufacturing, process control applications	Siemens, Phoenix Contact, B&R Automation
46	PROFINET IO	A real-time industrial Ethernet protocol for automation systems.	TCP/UDP: 34962-34964	IPsec encryption (Achieved through IPsec implementation)	Vulnerabilities in access control, authentication mechanisms	Industrial automation, process control applications	Siemens, Phoenix Contact, B&R Automation
47	PROFIsafe	A safety communication protocol for fail-safe automation systems.	TCP/UDP: 34962-34964	IPsec encryption (Achieved through IPsec implementation)	Vulnerabilities in access control, authentication mechanisms	Safety-critical applications, industrial automation	Siemens, Phoenix Contact, B&R Automation
48	S7Comm	A proprietary protocol used in Siemens S7-300 and S7-400 PLCs.	TCP: 102	Secure Authentication and TLS encryption	Vulnerabilities in authentication, data integrity	Industrial automation, control systems	Siemens, Schneider Electric, ABB
49	SNMP	A protocol for network management and monitoring of devices.	UDP: 161, 162	v3 has encryption	Vulnerabilities in authentication, data privacy	Network management, device monitoring and control	Cisco, Juniper Networks, Hewlett Packard Enterprise
50	Vnet	Yokogawa Proprietery protocol for Centum CS Controllers	UDP : 5313	Can use SSL/TSL encryption	Weak authentication , Data integrity, Dos	All sectors in Industrial Automation	Yokogawa
51	Vnet/IP	Yokogawa Proprietery protocol for Centum VP Controllers	TCP: 44818	Can use SSL/TSL encryption	Weak authentication , Data integrity, Dos	All sectors in Industrial Automation	Yokogawa
52	WirelessHART	A wireless communication protocol based on HART for industrial applications.	UDP: 5093	AES-128 encryption (Inherent encryption)	Vulnerable to jamming, unauthorized access	Wireless monitoring and control of industrial processes	Emerson, Honeywell, Siemens
53	WirelessMBus	A wireless communication protocol for utility metering applications.	TCP: 50000	AES-128 encryption (Inherent encryption)	Vulnerable to unauthorized access, data integrity issues	Wireless utility metering, remote meter reading	Kamstrup, Itron, Siemens
54	WISA	A wireless protocol for industrial automation and control.	UDP: 49200	Not Available (Plain Text)	Vulnerable to unauthorized access, data integrity issues	Wireless industrial control and monitoring, asset management	Endress+Hauser, Pepperl+Fuchs, ABB
55	WISA Wireless	A wireless protocol for communication in industrial automation.	UDP: 49200	Not Available (Plain Text)	Vulnerable to unauthorized access, data integrity issues	Wireless industrial control and monitoring, asset management	Endress+Hauser, Pepperl+Fuchs, ABB
56	Zigbee	A wireless communication protocol for low-power, low-data-rate IoT devices.	Various	AES encryption (Inherent encryption)	Vulnerabilities in authentication, data privacy	Home automation, smart lighting, wireless sensor networks	Philips, Texas Instruments, Silicon Labs
57	Zigbee IP	A variant of Zigbee protocol using IP networks for IoT applications.	Various	AES encryption (Inherent encryption)	Vulnerabilities in authentication, data privacy	IoT applications, wireless sensor networks with IP connectivity	Philips, Texas Instruments, Silicon Labs
58	Zigbee RF4CE	A variant of Zigbee protocol for remote control applications.	Various	AES encryption (Inherent encryption)	Vulnerabilities in authentication, data privacy	Remote controls, consumer electronics	Philips, Texas Instruments, Silicon Labs